

Web Novel Readers' Comment Sentiment Annotation Guidelines

Mohammad (2016) has discussed several sentence types that are particularly challenging to annotate for sentiment, including expressions of emotional states, multiple opinions within a single statement, and instances of sarcasm or ridicule. He proposed a semantic-based annotation questionnaire to address these complexities. Building on this foundation, we have adapted and expanded his framework specifically for reader comments on digital social reading platforms. Our adapted annotation guidelines account for the unique characteristics of web novel comments, including their conversational nature, reader-author dynamics, and genre-specific vocabulary. We then use this refined annotation framework to train BERT models, enabling more accurate sentiment classification for this distinct type of user-generated content.

The following are some common styles of reader comments

1. **Neutral Reporting:** Readers sometimes provide factual observations, clarifications, or descriptions of story events without expressing personal emotions or evaluations. These comments convey information rather than sentiment. Example: "china uses celsius so i'll assume that instead of fahrenheit." Such comments are annotated as neutral because they lack indicators of the reader's emotional state or attitude toward the content.
2. **Different sentiment towards different targets of opinion:** The reader may express opinions about multiple targets, and sentiment towards the different targets might be different. The targets may be different characters or events (for example, I love character A but not character B), or they may be different aspects of the same entity (for example, I love character A for his wisdom, but he is kind of cold blooded towards his enemy).
3. **Sarcasm and ridicule:** Sarcasm and ridicule are tricky from the perspective of assigning a single label of sentiment because they can often indicate a positive emotional state of the reader (pleasure from making fun of someone or something) even though they have a negative attitude towards someone or something. Readers of web novels frequently employ sarcastic comments when discussing characters or events, making valence determination ambiguous. During the annotation, we label the comment as positive or negative with a sublabel (Sarcasm/irony/mockery & Playful/humorous/joking) to indicate the emotional tendency. Consider the example: "Here comes the shiny scape-goat lol." This comment simultaneously jokes about the

character (negative sentiment toward the target) while expressing the reader's entertainment (positive emotional state). In this case, we label the comment as Positive with sublabel Playful/humorous/joking. The main label plus sublabel could help us better illustrate the complexity inherent in annotating web novel reader comments.

4. **Requests:** Many comments are requesting the update of new chapters from the author/translator. It can come off in both positive and negative emotions, such as "I can't wait for the next chapter..." and "why is it such a slow updates?"
5. **Quoting from the story or from other readers:** Quotations from the story or other comments pose annotation challenges because the commenter's stance toward the quoted content is often ambiguous. For example, a reader quoting "7 foot tall youngster" from the story uses a playful tone to highlight the contrast between the character's height and age. However, without additional context, the comment alone does not reveal whether the reader finds this amusing, absurd, or problematic, making sentiment determination difficult. In practice, we label these kinds of comments as neutral.
6. **Role-Play Comments:** Some readers commented by adopting the personas of characters. These comments blur the boundary between reader comments and imaginative participation in the narrative. For example, a reader might speak in that character's voice to express themselves or address another character that reflects the character's perspective or personality. For example, "MC: and you only tell me this now?" These comments are challenging to annotate because the sentiment expressed may represent the adopted character's attitude rather than the reader's genuine opinion. For annotation purposes, we label the sentiment conveyed in the role-played speech itself, regardless of the reader's actual attitude toward the character being mimicked. This approach treats the expressed emotion as the primary signal, even when delivered through a fictional persona.
7. **Writing Corrections:** Some readers identify grammatical, spelling, or translation errors in the text, offering corrections to authors or translators. For example: "rising-->raising." These comments are typically neutral in sentiment, as they focus on technical accuracy rather than expressing positive or negative attitudes toward the story itself.
8. **Rhetorical questions:** Rhetorical questions present annotation ambiguity as they can function either as neutral inquiries or as expressions of the reader's emotional state. For example, studies that focus on sarcasm define rhetorical questions as a subtype of sarcasm (Abbott, Ecker, Anand and Walker, 2016). Consider the comment

"Wouldn't that also add impurity trash in the meridians!!??" This comment can be interpreted as negative sentiment expressing skepticism or criticism of the story's internal logic, and the presence of multiple punctuation marks amplifies the emotion, making these comments particularly challenging to classify with a single sentiment label. In this case, we can add a sublabel for the sentiment. However, examples like "uk its kinda strange they have jeans rights?" could be just neutral.

9. **Punctuation & Typographic Features:** Punctuation marks and typographic emphasis are not merely grammatical tools; they are critical indicators of sentiment intensity, irony, and hidden meaning. Annotators must treat these non-verbal cues as behavioral modifiers that can amplify, mitigate, or reverse the sentiment of the text.

9.1 Rhetorical Questions

Rhetorical questions differ from standard questions as they possess the syntactic form of a question but serve the function of a statement.

- **Definition & Function:** As Ranganath et al. (2016) mentioned, rhetorical questions generally do not seek information; instead, they serve the function of a statement and are used to imply a message implicitly from the context. They often function as indirect speech acts.
- **Negative Bias & Persuasion:** Ranganath et al. (2016) further noted that rhetorical questions are widely employed persuasion tools. In social media contexts, they are frequently used to express thoughts or criticism rather than genuine inquiry.

E.g., "Wouldn't that be sth!!??"

Annotation Rule: Do not annotate as a Neutral inquiry. Identify the underlying assertion (likely Negative criticism).

9.2 Indicators of Sarcasm and Irony

Sarcasm is a nuanced form of language where individuals state the opposite of what is implied. Rajadesingan et al. (2015) highlighted that because written text lacks the vocal cues of speech (such as pitch, intensity, and tempo) , users rely on specific writing styles to compensate for these missing non-verbal cues.

- **Ellipses (...):**
Rajadesingan et al. (2015) identified punctuation marks like ellipses as non-verbal cues crucial for sarcasm deliverance. Specifically, ellipses (...) are often used to indicate a pause, In feedback, this often signals hesitation, speechless frustration, or a trailing thought that implies a negative critique.
-Label: check the negative (speechless, disappointed).
- **Asterisks (*) & Quotation Marks:**
According to Rajadesingan et al. (2015), users may use asterisks (*) to indicate emphasis. When used with positive words in a negative context (e.g., *genius*), this signals irony.
- **Over-done Punctuation (!!!):**

Rajadesingan et al. (2015) also noted that exclamation marks (!!!) are sometimes "over-done" to indicate sarcasm. If the punctuation intensity conflicts with the context, it is a marker of irony.

9.3 Sentiment Intensity Amplifiers (Typographic Features)

Users employ typographic strategies to replace vocal stress and loudness found in spoken language.

- Repeated Punctuation (!! , ?? !?):
Multiple question marks or exclamation marks serve to amplify the emotion expressed in the text.
- Repeated Characters (Elongation):
Rajadesingan et al. (2015) observed that users often repeat letters in words (e.g., "sooooo", "awesomEEEE") to stress and over-emphasize certain parts of the tweet. Example: "This is sooooo good" should be rated higher in positive intensity than "This is so good".

Annotation Rule: Treat words with repeated characters (3 or more) as indicators of high-intensity sentiment.

- Capitalization (All-Caps):
Rajadesingan et al. (2015) explained that capitalization is often used to emphasize changes in tone, as if reading out loud. Users may capitalize words or specific parts-of-speech to exaggerate or to vent their frustration.

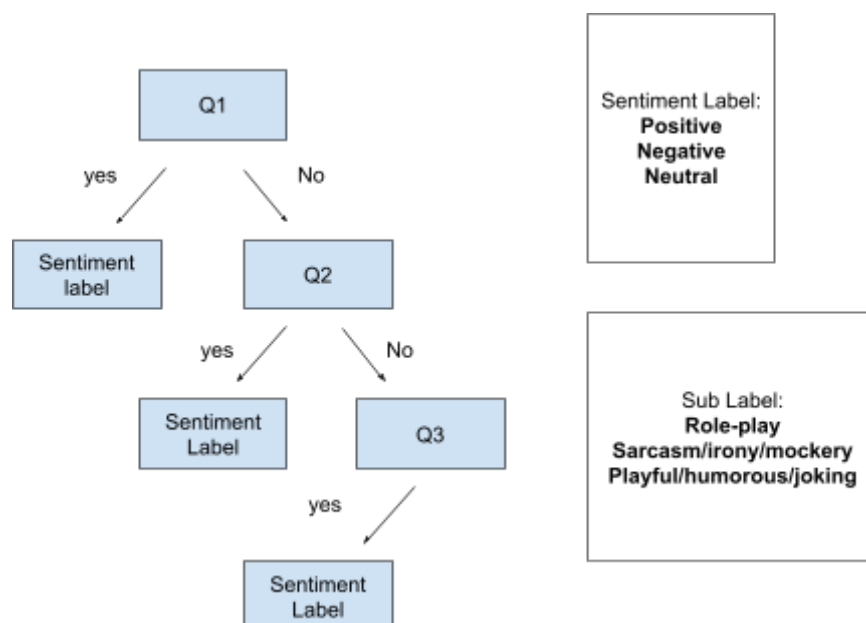
Annotation Rule: Treat fully capitalized words as Strong sentiment indicators.

10. Comments with emoji/emoticon: Emoticons have proved crucial in the automated sentiment classification of informal texts. It allows the author to express feelings, moods and emotions, and augments a written message with non-verbal elements. It helps to draw the reader's attention, and enhances and improves the understanding of the message. When annotating comments that contain emojis or emoticons, treat these visual elements as modifiers rather than overrides of sentiment. The text content should take precedence in determining the primary sentiment, while emojis serve to shift intensity or add nuance to the overall tone. For example, a comment like "I love this! 😍", the emojis can intensify the sentiment of the text. When an emoji clearly contradicts the literal text in a way that signals sarcasm or irony, for example "wow this exactly what I want 🤔" annotate based on the likely intended sentiment, but note this as a special case. If the emoji's function is unclear or purely decorative, default to the text-based sentiment. This approach ensures consistency while capturing the emotional complexity that emojis add to digital communication.

SEMANTIC-ROLE BASED SENTIMENT Decision Tree for Web Novel

In our classification of the sentiment of reader comments, we only give one label to the whole comment (which could consist of an incomplete sentence or multiple sentences), so it is important to decide which is the primary sentiment of the comment. We design the following questions to help classify the sentiment based on Mohamad's guidelines.

Decision Tree



Q1. From reading the text, the speaker's emotional state can best be described as:

• **Positive state:** there is an explicit or implicit clue in the text suggesting that the reader is (predominantly) in a positive state, i.e., happy, admiring, relaxed, forgiving, appreciative, thankful, excited, **playful, humorous, joking**, optimistic, or inspired, etc.

Example 1: 麻雀虽小, 但是这种生物从没有被人类成功驯养过, 是大自然里最热爱自由的生物

Example 2: (ଢ଼ଢ଼)hiahiahia脱缰的野驴笑死我了

Example 3: Smart people instead of dramatic ones, Yay

• **Negative state:** there is an explicit or implicit clue in the text suggesting that the reader is (predominantly) in a negative state, i.e., sad, angry, anxious, violent, critical, disappointed, pessimistic, **sarcasm, irony, or mockery**, etc.

Example 1: 香菜可太难吃了。吃的我都恶心

Example 2: 麻皮，混字数啊，还能要点碧莲不，来自书友的负面情绪666

Example 3: Ahh it's his fault he should have just died

Example 4: Could have just done that to begin with.

Example 5: if you're going to give a reason at least give a decent one, like it just feels right in my hands I feel like I can conquer the world and sweep aside floods and conquer nations with a spear

• **Neutral state:** there is no explicit or implicit indicator of the reader's emotional state. Or the reader just stating factual knowledge without any emotional tendency.

Example 1: 黑面包的面粉里面纯面粉含量太低，不过度发酵根本发不起来。然而，过度发酵的面团就是发酸的。

Example 2: 麻雀猴子吃素？实际上纯粹的食草动物是很少很少的，基本都是杂食。食肉动物也会吃草。

Example 3: I'm guessing "Cat" was his brother?

Example 4: Yes, that all dangerous fruit knife. Cousin to the butter knife, the fruit knife is not as widely known or used, but far deadlier.

If there are both positive and negative, or mixed, feelings: there is an explicit or implicit clue in the text suggesting that the speaker is experiencing both positive and negative feelings.

After classifying the speaker's emotional state, GO TO Q2

Q2. From reading the text, identify the entity towards which opinion is being expressed or the entity towards which the speaker's attitude can be determined.

This entity is usually a character, event, setting, author, readers, or some such entity. We will call this the **PRIMARY TARGET OF OPINION (PTO)**.

If there is **more than one target of opinion**, then select that target **towards which sentiment is stronger**.

Example 1: 骆馨雨来交好小栗, 实属正常, 我觉的这里就是骆馨雨是咱们大家的化身, 小槿, 小栗, 许显楚都有实力, 又有肘子护佑, 骆馨雨就是咱们大家, 为了生存, 为了工作养家糊口, 无奈得奉承上司, 小栗你别忘记, 你还是骆馨雨雇你的时间内, 那些私军是不可靠了, 他们都色眼对骆馨雨了, **Positive**

The sentence is expressing emotions towards characters as PTO. Many character names are mentioned, e.g. 骆馨雨 and 小栗. However, the stronger emotion is more towards the 骆馨雨(PTO) as it is being mentioned the most.

Example 2: 骆馨雨挺不错的啊, 人家所做一切都是为了保命, 而且所用的东西也是自己有的。没必要为了强行打击女人来取悦读者。**Negative**

The first sentence expresses emotions towards characters as PTO. And the second sentence expresses emotions towards the author as PTO. However, 骆馨雨 is the SUBJECT being discussed, but the ADDRESSEE of the argument is the author/critics, so the stronger emotion is more towards the author (PTO).

If there is no explicit opinion of the speaker, then go to Q3

Q3: What best describes the speaker's attitude, evaluation, or judgment towards the primary target of opinion (PTO)? If the whole text is trying to make comments mimic the voice/opinion of the character (etc. PERSON mentioned) and there is no indication of the speaker's attitude, then decide the emotion based on the SPEECH content.

Example 1: 黑医任小粟: 六元你还年轻啊, 这跟捞鱼一个样, 小病不治, 等小病长大了就能捞一票大的了 **Positive**

In this example, there's no explicit expression of the author's opinion. Then adopt the expression of 任小粟 as the agent of the author. And the expression of 任小粟 is positive.

Example 2: The moral of this chapter: never underestimate the MC **Positive**

References

Abbott, R., Ecker, B., Anand, P., & Walker, M. (2016). Internet Argument Corpus 2.0: An SQL schema for Dialogic Social Media and the Corpora to go with it. In N. Calzolari, K. Choukri, T. Declerck, S. Goggi, M. Grobelnik, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, & S. Piperidis (Eds), *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)* (pp. 4445–4452). European Language Resources Association (ELRA). <https://aclanthology.org/L16-1704/>

Mohammad, S. (2016). A Practical Guide to Sentiment Annotation: Challenges and Solutions. In A. Balahur, E. van der Goot, P. Vossen, & A. Montoyo (Eds), *Proceedings of the 7th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis* (pp. 174–179). Association for Computational Linguistics. <https://doi.org/10.18653/v1/W16-0429>

Ranganath, S., Hu, X., Tang, J., Wang, S., & Liu, H. (2021). Identifying Rhetorical Questions in Social Media. *Proceedings of the International AAAI Conference on Web and Social Media*, 10(1), 667-670. <https://doi.org/10.1609/icwsm.v10i1.14771>

Rajadesingan, A., Zafarani, R., & Liu, H. (2015). Sarcasm Detection on Twitter: A Behavioral Modeling Approach. Proceedings of the Eighth ACM International Conference on Web Search and Data Mining, 97–106. <https://doi.org/10.1145/2684822.2685316>